FOCUS 3 • PHYSICAL CHEMISTRY • 物理化学

The Second and Third Laws of Thermodynamics

热力学第二、第三定律

Entropy, spontaneity, Gibbs energy and the Third Law

REFERENCE · 对照表

Key concepts · 重要概念

Key concepts · 中文对照 · phonetics

entropy
熵 /ˈentrəpi/
second law
第二定律 /ˈsekənd lɔː/
spontaneous
自发 /spɒnˈteɪniəs/
reversible
可逆 /rɪˈvɜːsəbl/
Carnot cycle
卡诺循环 /kɑːˈnoʊ ˈsaɪkl/
Gibbs energy G
Gibbs能 G /gɪbz ˈenədʒi/
Helmholtz energy A
Helmholtz能 A /ˈhelmhoʊlts/
absolute entropy
绝对熵 /ˈæbsəluːt ˈentrəpi/
heat capacity
热容 /hiːt kəˈpæsəti/
phase transition
相变 /feɪz trænˈzɪʃn/
Maxwell relations
麦克斯韦关系 /ˈmækswel/
perfect crystal
完美晶体 /ˈpɜːfɪkt ˈkrɪstl/

What we will cover

- 3A Entropy and the Second Law
  熵与第二定律
- 3B Entropy changes in processes
  过程熵变
- 3C Measuring entropy · Third Law
  熵的测量与第三定律
- 3D Helmholtz and Gibbs energies
  Helmholtz与Gibbs能
- 3E Combining the First and Second Laws
  两定律结合

· Entropy and Gibbs energy decide whether a change is spontaneous.

Learning outcomes

- State the Second Law and define entropy.
  第二定律
- Compute ΔS for expansion, phase change and heating.
  熵变
- Use the Third Law to set absolute entropies.
  第三定律
- Define G and use ΔG for spontaneity.
  Gibbs能
- Combine the First and Second Laws.
  结合两定律

· By the end you can predict the direction of a change.

FOCUS 3 · WARM-UP

Why do some things just happen?

Think about it · 想一想

A gas spreads into a vacuum; a hot cup cools. What microscopic 'direction' do these share, and how do we measure it?

Think about the dispersal of energy and matter — that quantity is entropy.

TOPIC 3A

Entropy

熵

➤ Why do you need to know this material?

Entropy is the concept on which almost all applications of thermodynamics in chemistry are based: it explains why some physical transformations and chemical reactions are spontaneous and others are not.

➤ What is the key idea?

The change in entropy of a system can be calculated from the heat transferred to it reversibly; a spontaneous process in an isolated system is accompanied by an increase in entropy.

1. The Second Law · 第二定律
2. Definition of entropy · 熵的定义
3. Entropy as a state function · 熵是状态函数

Spontaneous change自发变化

- Some changes happen naturally; some do not.
- The direction is determined by the manner in which energy and matter are distributed.
- The localization of random, disorderly motion as directed, orderly motion is so unlikely.

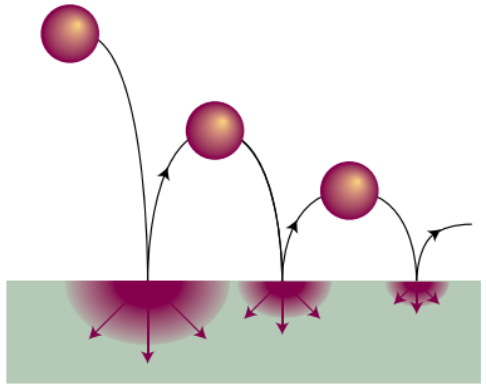


Figure 3A.1 The direction of spontaneous change for a ball bouncing on a floor. On each bounce some of its energy is degraded into the thermal motion of the atoms of the floor, and that energy then disperses. The reverse process, a ball rising from the floor as a result of acquiring energy from the thermal motion of the atoms in the floor, has never been observed to take place.

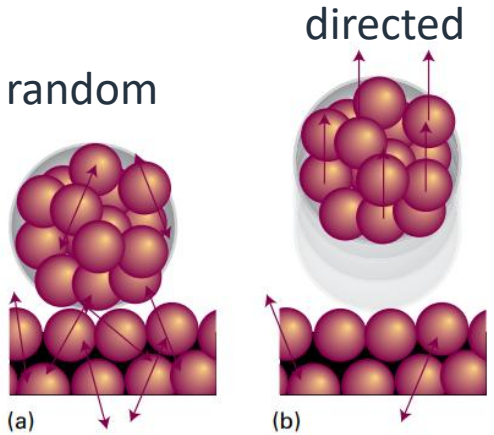


Figure 3A.2 (a) A ball resting on a warm surface; the atoms are undergoing thermal motion (vibration, in this instance), as indicated by the arrows. (b) For the ball to fly upwards, some of the random vibrational motion would have to change into coordinated, directed motion. Such a conversion is highly improbable.

· 'Spontaneous' means a natural tendency, not necessarily fast.

Second Law of thermodynamics

- look for the direction of change that leads to the dispersal of energy
- Matter also has a tendency to disperse分散.
- Kelvin: No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.
只产生“从单一热源吸取热量并将其完全转变为功”这个唯一结果的过程是不存在的

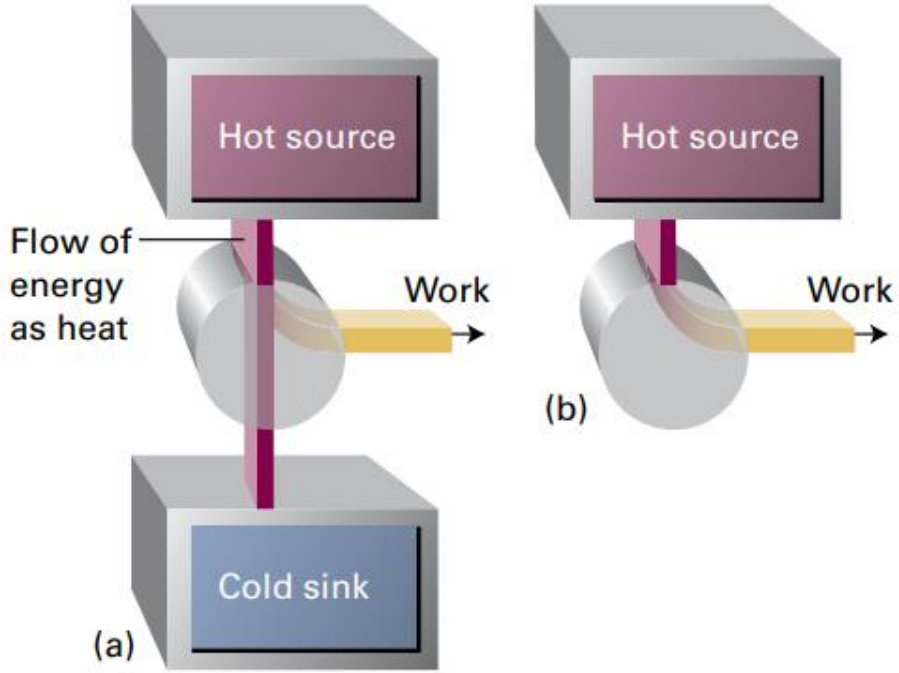


Figure 3A.3 (a) A heat engine is a device in which energy is extracted from a hot reservoir (the hot source) as heat and then some of that energy is converted into work and the rest discarded into a cold reservoir (the cold sink) as heat. (b) The Kelvin statement of the Second Law denies the possibility of the process illustrated here, in which heat is changed completely into work, there being no other change.

Second Law of thermodynamics

- Rudolf Clausius : Heat does not flow spontaneously from a cool body to a hotter body.热量不会自发地从低温物体流向高温物体
- To achieve the transfer of heat to a hotter body, it is necessary to do work on the system, as in a refrigerator.
- Clausius statement is logically equivalent to the Kelvin statement

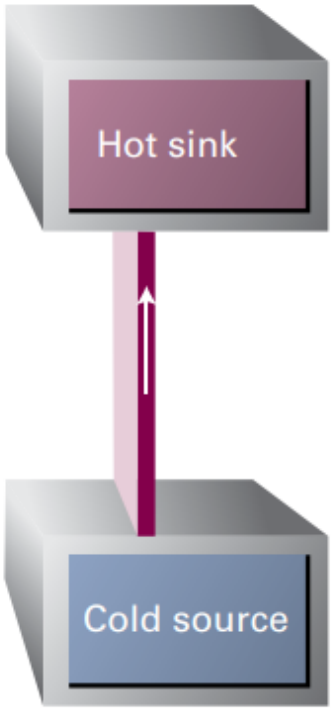


Figure 3A.4 According to the Clausius statement of the Second Law, the process shown here, in which energy as heat migrates from a cool source to a hot sink, does not take place spontaneously. The process is not in conflict with the First Law because energy is conserved.

不可逆过程中，除了热传递带来的混乱度增加，还有耗散效应（摩擦、涡流、自发扩散等），会额外产生熵。如果直接拿实际不可逆过程的热去算，得到的数值会偏小，不等于真实的熵差。可逆过程中系统和环境始终温差无限小，没有额外耗散产生的熵

The definition of entropy

- A change in the extent to which energy is dispersed in a disorderly way depends on how much energy is transferred as heat, not as work.
- The thermodynamic definition of entropy is based on the expression

$$dS = \frac{dq_{rev}}{T}$$

Entropy change [definition] (3A.1a)

- q_{rev} is the energy transferred as heat reversibly to the system at the absolute temperature T. For a measurable change between two states i and f,

$$\Delta S = \int_i^f \frac{dq_{rev}}{T} \qquad \Delta S \geq \int \frac{dq}{T}$$

- the units of entropy are joules per kelvin (J K^{-1}). Entropy is an extensive property.
Molar entropy, Sm = S/n, (J K^{-1} mol^{-1}); is an intensive property

· Entropy – reversible heat change

Example 3A.1

$$\Delta S = \int_i^f \frac{dq_{rev}}{T}$$

- Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f.
- Collect your thoughts The definition of entropy change in eqn 3A.1b instructs you to find the energy supplied as heat for a reversible path between the stated initial and final states regardless of the actual manner in which the process takes place. The process is isothermal, so T can be treated as a constant and taken outside the integral in eqn 3A.1b. Moreover, because the internal energy of a perfect gas is independent of its volume (Topic 2A), $\Delta U = 0$ for the expansion. Then, because $\Delta U = q + w$, it follows that $q = -w$, and therefore that $q_{rev} = -w_{rev}$. The work of reversible isothermal expansion is calculated in Topic 2A. Finally, calculate the change in molar entropy from $\Delta S_m = \Delta S/n$.

Example 3A.1

$$\Delta S = \int_i^f \frac{dq_{rev}}{T}$$

- Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f.
-

The solution The temperature is constant, so eqn 3A.1b becomes

$$\Delta S = \frac{1}{T}\int_i^f dq_{rev} = \frac{q_{rev}}{T}$$

From Topic 2A the reversible work in an isothermal expansion is $w_{rev} = -nRT\ln(V_f/V_i)$, hence $q_{rev} = nRT\ln(V_f/V_i)$. It follows, after dividing q_{rev} by T, that

$$\Delta S = nR\ln\frac{V_f}{V_i} \quad \text{and} \quad \Delta S_m = R\ln\frac{V_f}{V_i}$$

Self-test 3A.1

$$\Delta S = \int_i^f \frac{dq_{rev}}{T} \qquad \Delta S = nR \ln \frac{V_f}{V_i}$$

- Calculate the change in entropy when the pressure of a fixed amount of perfect gas is changed isothermally from p_i to p_f. What is the origin of this change?

$$p_i V_i = p_f V_f, \quad \frac{V_f}{V_i} = \frac{p_i}{p_f}$$

$$\Delta S = nR \ln \frac{p_i}{p_f}$$

Origin of entropy change: Molecular kinetic energy remains constant at constant temperature; pressure change leads to volume change.

$$dS = \frac{dq_{rev}}{T}$$ Entropy change [definition] (3A.1a)

The change in entropy of the surroundings

- The surroundings consist of a reservoir of constant volume, so the energy supplied to them by heating can be identified with the change in the internal energy of the surroundings, $dU_{sur}=dq_{sur}$.

$$dS_{sur} = \frac{dq_{sur}}{T_{sur}}$$ Entropy change of the surroundings (3A.2a)

- the temperature of the surroundings is constant

$$\Delta S_{sur} = \frac{q_{sur}}{T_{sur}}$$ (3A.2b)

- for any adiabatic change, qsur = 0, so

$$\Delta S_{sur} = 0$$ Adiabatic change (3A.3)

Example 3A.1

$$\Delta S = \int_i^f \frac{dq_{rev}}{T}$$

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$$\Delta S_{sur} = \frac{q_{sur}}{T_{sur}} \qquad (3A.2b)$$

The change in entropy of the surroundings

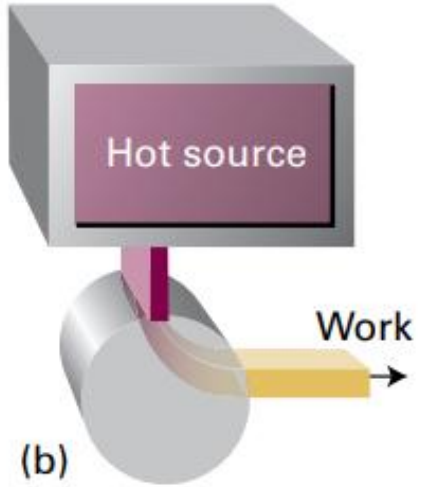
Brief illustration 3A.1

To calculate the entropy change in the surroundings when 1.00 mol $H_2O(l)$ is formed from its elements under standard conditions at 298 K, use $\Delta_f H^{\ominus} = -286\,kJ\,mol^{-1}$ from Table 2C.4. The energy released as heat from the system is supplied to the surroundings, so $q_{sur} = +286\,kJ$. Therefore,

$$\Delta S_{sur} = \frac{2.86\times10^{5}\,J}{298\,K} = +960\,J\,K^{-1}$$

This strongly exothermic reaction results in an increase in the entropy of the surroundings as energy is released as heat into them.

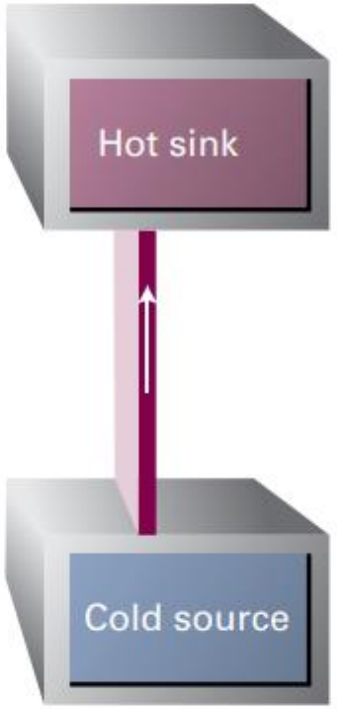
Second Law of thermodynamics



$\Delta S\downarrow$

ΔS unchanged

$\Delta S_{tot}<0$, not spontaneous



$q_{high}=q_{cold}$

Boltzmann distribution

- The total number of molecules with a particular energy due to translation平动, rotation转动, vibration振动, and its electronic state is called the ‘population布居’

- Most molecules are found in the lowest energy state, and higher energy states are occupied by progressively fewer molecules. The Boltzmann distribution gives the population, Ni, of any energy state in terms of the energy of the state, εi, and the absolute temperature, T.

$$N_i \propto e^{-\varepsilon_i/kT}$$

- k is Boltzmann’s constant

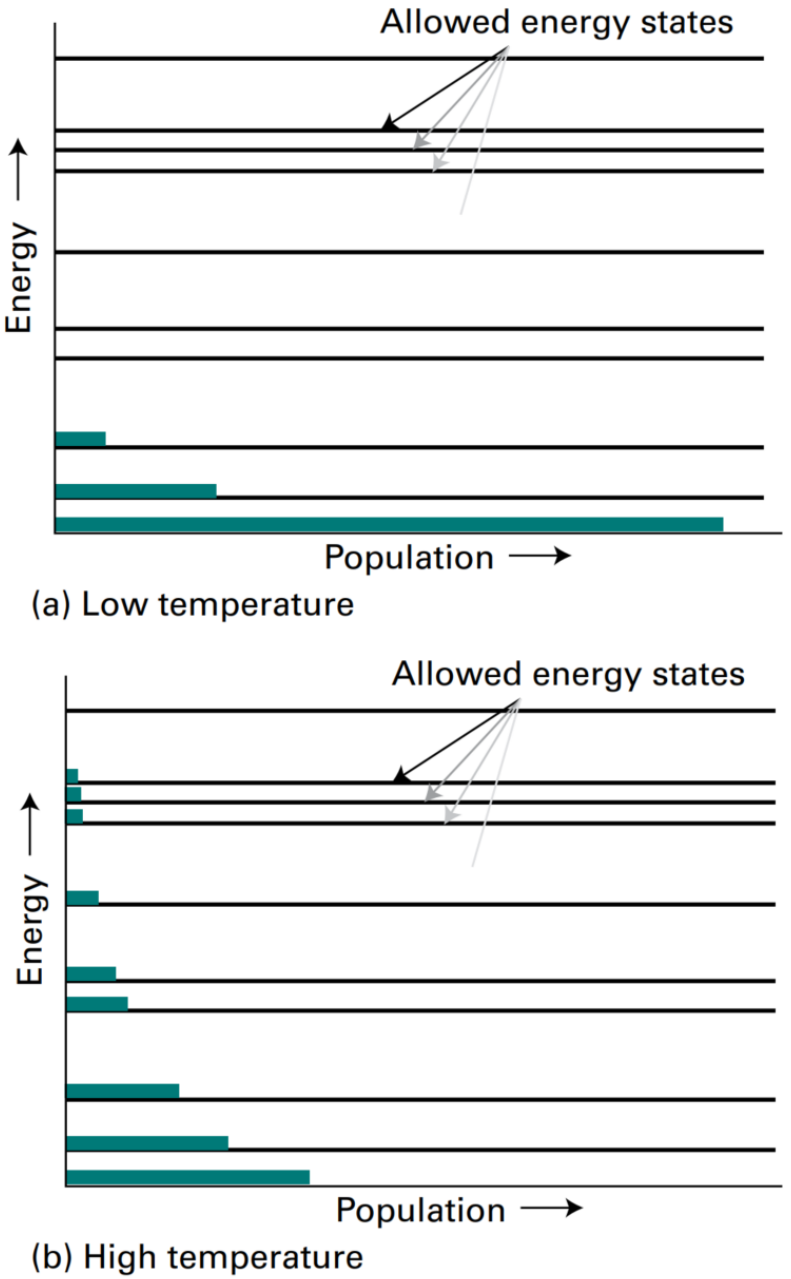


Figure 2 The relative populations of states at (a) low, (b) high temperature according to the Boltzmann distribution.

The statistical definition of entropy

- Boltzmann proposed that there is a link between the spread of molecules over the available energy states and the entropy

statistical entropy
统计熵

$$S = k \ln \mathcal{W}$$

Boltzmann formula for the entropy (3A.4)

- W is the number of microstates, the number of ways in which the molecules of a system can be distributed over the energy states for a specified total energy
- If all the molecules are in one energy state there is only one way of achieving this distribution, so W = 1 and, because ln 1 = 0, it follows that S = 0.

$$S = k \ln \mathcal{W}$$ Boltzmann formula for the entropy (3A.4)

The statistical definition of entropy

- Increasing the energy of the system by the transfer of heat makes more states accessible, but given that very many states are already occupied the proportionate change in W is small

- For a system at a low temperature fewer states are occupied, and so the transfer of the same energy results in a proportionately larger increase in the number of accessible states, and hence a larger increase in W.

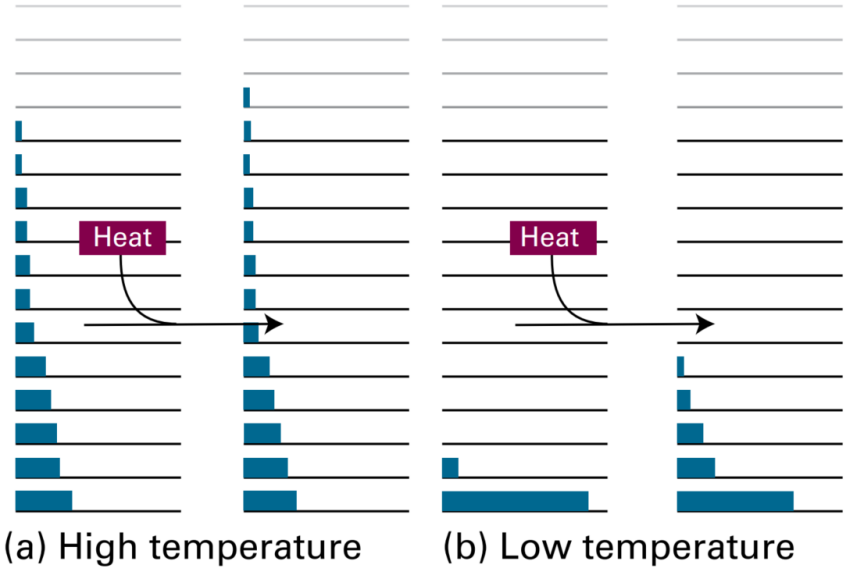


Figure 3A.6 The supply of energy as heat to the system results in the molecules moving to higher energy states, so increasing the number of microstates and hence the entropy. The increase in the entropy is smaller for (a) a system at a high temperature than (b) one at a low temperature because initially the number of occupied states is greater.

Entropy as a state function (Carnot)

- Entropy is a state function. the integral of dS between any two states is independent of the path between them

$$\oint dS = \oint \frac{dq_{rev}}{T} = 0 \qquad (3A.5)$$

- The symbol $\oint$ denotes integration around a closed path沿循环途径积分. There are three steps in the argument:

1. First, to show that eqn 3A.5 is true for a special cycle (a 'Carnot cycle') involving a perfect gas.
2. Then to show that the result is true whatever the working substance.
3. Finally, to show that the result is true for any cycle.

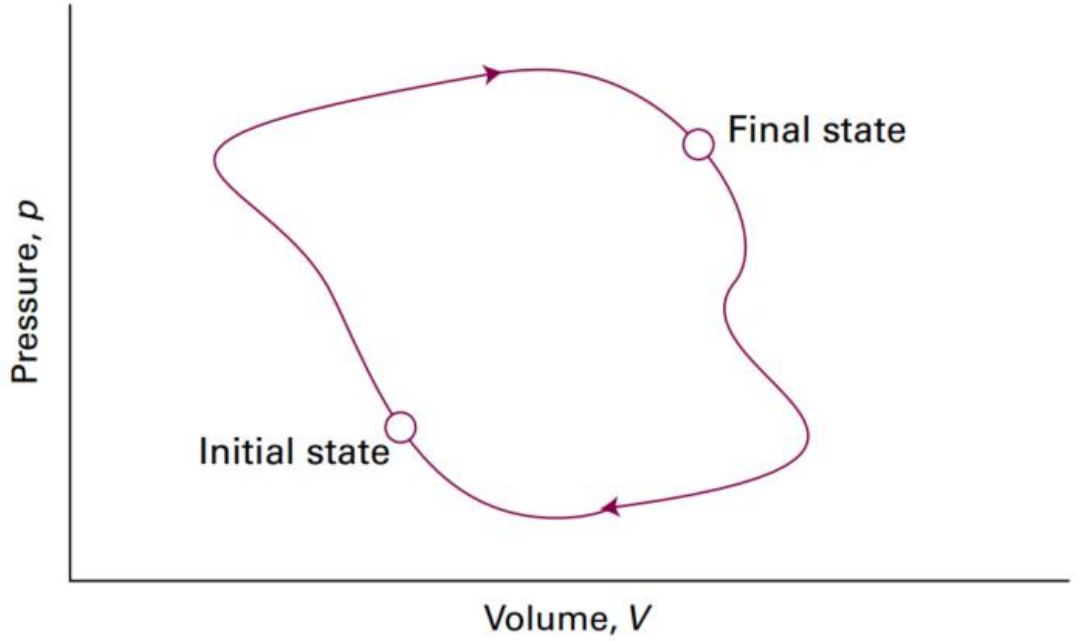


Figure 3A.7 In a thermodynamic cycle, the overall change in a state function (from the initial state to the final state and then back to the initial state again) is zero.

French engineer
Sadi Carnot



Entropy as a state function (Carnot卡诺)

- A Carnot cycle卡诺循环, consists of four reversible stages in which a gas (the working substance) is either expanded or compressed in various ways; in two of the stages energy as heat is transferred to or from a hot source or a cold sink
- A Carnot cycle is a reversible engine cycle (isothermal + adiabatic).

- The four stages return the system to its start.
- $\oint dq_{rev}/T = 0$, so $\Delta S = 0$ for a round trip.
  环路熵变为零

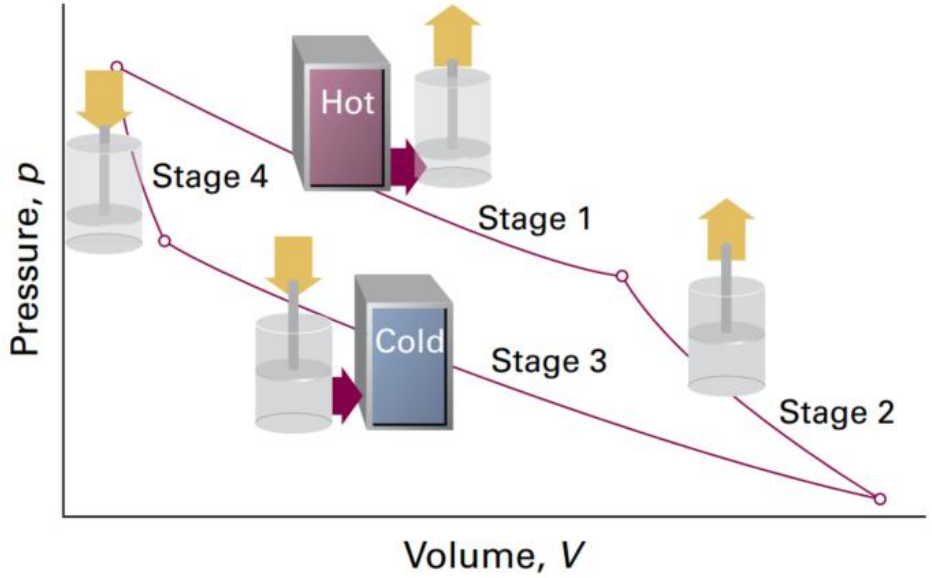


Figure 3A.8 The four stages which make up the Carnot cycle. In stage 1 the gas (the working substance) is in thermal contact with the hot reservoir, and in stage 3 contact is with the cold reservoir; both stages are isothermal. Stages 2 and 4 are adiabatic, with the gas isolated from both reservoirs.

· The Carnot cycle proves that entropy is a state function.

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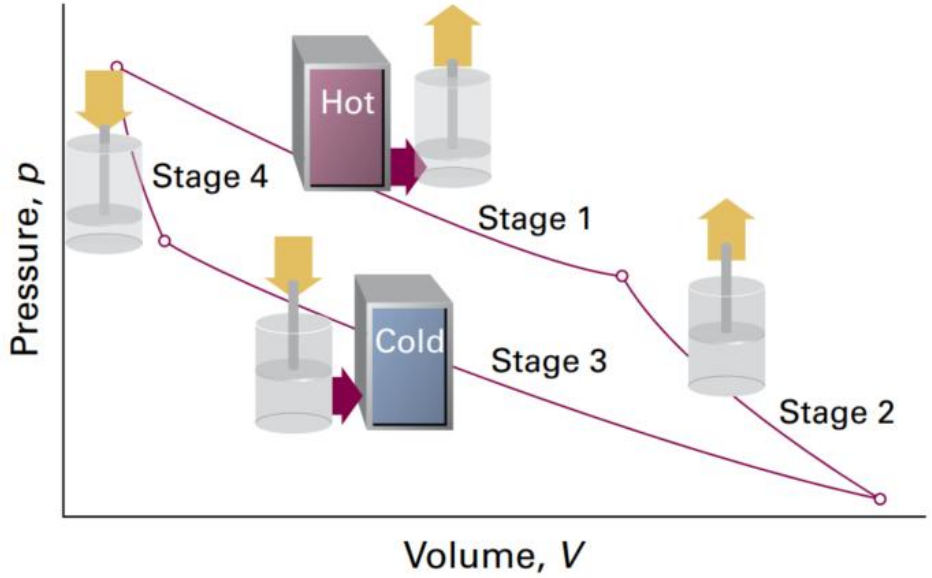


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· The Carnot cycle proves that entropy is a state function.

The Carnot cycle on a p–V diagram

- 1. The gas is placed in thermal contact with the hot source at temperature T_h and undergoes reversible isothermal expansion from A to B; the entropy change is q_h/T_h, where qh is the energy supplied to the system as heat from the hot source.

2. The gas undergoes reversible adiabatic expansion from B to C. No energy leaves the system as heat, so the change in entropy is zero. The expansion is carried on until the temperature of the gas falls from T_h to T_c, the temperature of the cold sink.

3. The gas undergoes a reversible isothermal compression from C to D at T_c. The change in entropy of the system is q_c/T_c; in this expression q_c is negative.

· The adiabats are steeper陡峭的 than the isotherms.

4. Finally, the gas then undergoes reversible adiabatic compression from D to A such that the final temperature is T_h. No energy enters the system as heat, so the change in entropy is zero.

$$\oint dS = \frac{q_h}{T_h} + \frac{q_c}{T_c}$$

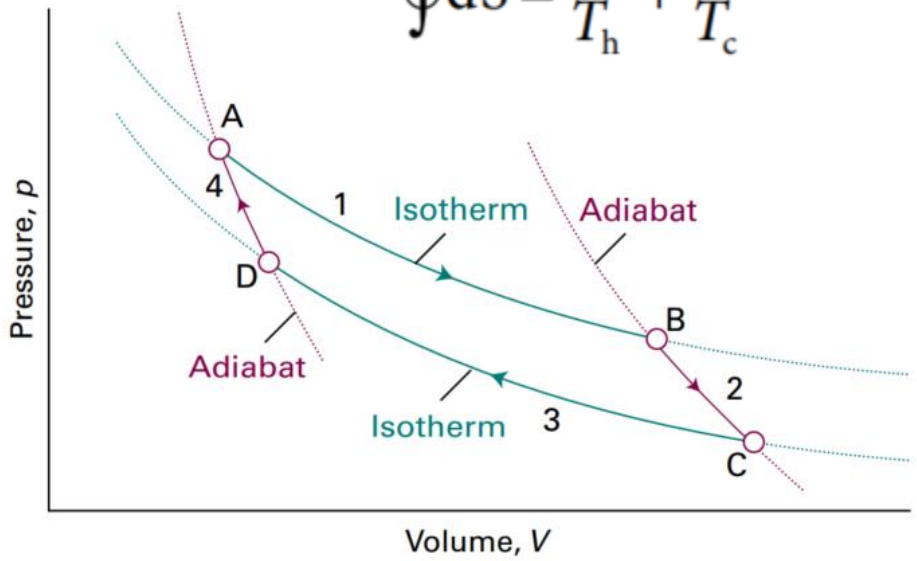


Figure 3A.9 The basic structure of a Carnot cycle. Stage 1 is the isothermal reversible expansion at the temperature T_h. Stage 2 is a reversible adiabatic expansion in which the temperature falls from T_h to T_c. Stage 3 is an isothermal reversible compression at T_c. Stage 4 is an adiabatic reversible compression, which restores the system to its initial state.

The Carnot cycle for perfect gas

How is that done? 3A.1 Showing that the entropy is a state function for a perfect gas

First, you need to note that a reversible adiabatic expansion (stage 2 in Fig. 3A.9) takes the system from T_h to T_c. You can then use the properties of such an expansion, specifically VT^c = constant (Topic 2E), to relate the two volumes at the start and end of the expansion. You also need to note that energy as heat is transferred by reversible isothermal processes (stages 1 and 3) and, as derived in *Example* 3A.1, for a perfect gas

$$\overbrace{q_h = nRT_h \ln \frac{V_B}{V_A}}^{\text{stage 1}} \qquad \overbrace{q_c = nRT_c \ln \frac{V_D}{V_C}}^{\text{stage 3}}$$

Step 1 *Relate the volumes in the adiabatic expansions*

For a reversible adiabatic process the temperature and volume are related by VT^c = constant (Topic 2E). Therefore

for the path D to A (stage 4): $V_A T_h^c = V_D T_c^c$

for the path B to C (stage 2): $V_C T_c^c = V_B T_h^c$

Multiplication of the first of these expressions by the second gives

$$V_A V_C T_h^c T_c^c = V_D V_B T_h^c T_c^c$$

which, on cancellation of the temperatures, simplifies to

$$\frac{V_D}{V_C} = \frac{V_A}{V_B}$$

The Carnot cycle for perfect gas

Step 2 *Establish the relation between the two heat transfers*

You can now use this relation to write an expression for energy discarded as heat to the cold sink in terms of V_A and V_B

$$q_c = nRT_c \ln\frac{V_D}{V_C} = nRT_c \ln\frac{V_A}{V_B} = -nRT_c \ln\frac{V_B}{V_A}$$

It follows that

$$\frac{q_h}{q_c} = \frac{nRT_h \ln(V_B/V_A)}{-nRT_c \ln(V_B/V_A)} = -\frac{T_h}{T_c}$$

Note that q_h is negative (heat is withdrawn from the hot source) and q_c is positive (heat is deposited in the cold sink), so their ratio is negative. This expression can be rearranged into

$$\frac{q_h}{T_h} + \frac{q_c}{T_c} = 0 \qquad (3A.6)$$

Because the total change in entropy around the cycle is $q_h/T_h + q_c/T_c$, it follows immediately from eqn 3A.6 that, for a perfect gas, this entropy change is zero.

The Carnot cycle for perfect gas

Brief illustration 3A.2

The Carnot cycle can be regarded as a representation of the changes taking place in a heat engine in which part of the energy extracted as heat from the hot reservoir is converted into work. Consider an engine running in accord with the Carnot cycle, and in which 100 J of energy is withdrawn from the hot source ($q_h = -100\,J$) at 500 K. Some of this energy is used to do work and the remainder is deposited in the cold sink at 300 K. According to eqn 3A.6, the heat deposited is

$$q_c = -q_h \times \frac{T_c}{T_h} = -(-100\,J) \times \frac{300\,K}{500\,K} = +60\,J$$

This value implies that 40 J was used to do work.

$$\frac{q_h}{T_h} + \frac{q_c}{T_c} = 0$$

Efficiency of heat engine

- The efficiency, η (eta), of a heat engine

$$\eta = \frac{\text{work performed}}{\text{heat absorbed from hot source}} = \frac{|w|}{|q_h|}$$ Efficiency [definition] (3A.7)

- All efficiencies are positive numbers.
- The definition implies that the greater the work output for a given supply of heat from the hot source, the greater is the efficiency of the engine
- The energy supplied as work by the engine is the difference between the energy supplied as heat by the hot source and that returned to the cold sink

$$\eta = \frac{|q_h| - |q_c|}{|q_h|} = 1 - \frac{|q_c|}{|q_h|} \quad (3A.8)$$

It then follows from eqn 3A.6, written as $|q_c|/|q_h| = T_c/T_h$ that

$$\eta = 1 - \frac{T_c}{T_h}$$ Carnot efficiency (3A.9)

Efficiency of heat engine

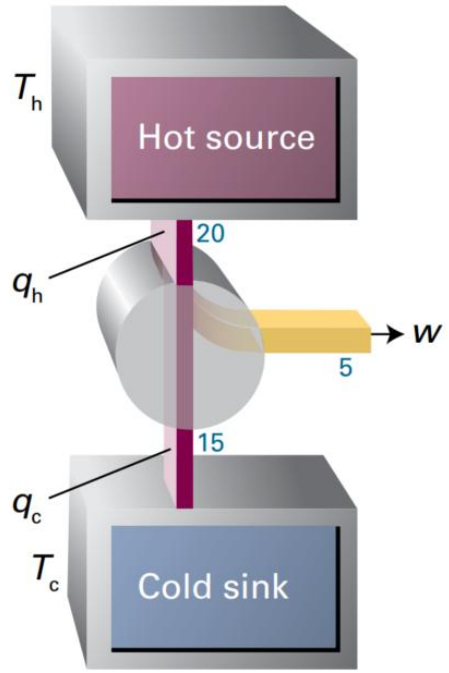


Figure 3A.10 In a heat engine, an energy q_h (for example, $|q_h| = 20\,kJ$) is extracted as heat from the hot source and q_c is discarded into the cold sink (for example, $|q_c| = 15\,kJ$). The work done by the engine is equal to $|q_h| - |q_c|$ (e.g. $20\,kJ - 15\,kJ = 5\,kJ$).

Brief illustration 3A.3

A certain power station operates with superheated steam at 300 °C ($T_h = 573\,K$) and discharges the waste heat into the environment at 20 °C ($T_c = 293\,K$). The theoretical efficiency is therefore

$$\eta = 1 - \frac{293\,K}{573\,K} = 0.489$$

or 48.9 per cent. In practice, there are other losses due to mechanical friction and the fact that the turbines do not operate reversibly.

Efficiency of heat engine

- The Second Law of thermodynamics implies that all reversible engines have the same efficiency regardless of their construction.
- Suppose two reversible engines are coupled together and run between the same hot source and cold sink
- 设想有两台可逆热机，接在同一个高温热源、同一个低温热源上。先假设：A 热机效率 > B 热机效率。
- 让 A 正常运转，对外做功；因为 B 是可逆机器，可以反过来当成制冷机，消耗功，把热量从低温送到高温。我们用 A 输出的功去带动反向的 B。算整体效果：最后只从高温热源拿热，全部变成功，低温热源热量没变。这就违反热力学第二定律开尔文说法：不能只靠单一热源吸热，全部变成功，不带来别的变化。所以我们一开始 “A 效率更高” 这个假设是错的。同理也不能 B>A。一句话大白话：如果两个可逆热机效率不一样，拼在一起就能造出第二类永动机，而热力学第二定律禁止第二类永动机，所以它们效率必须相等。

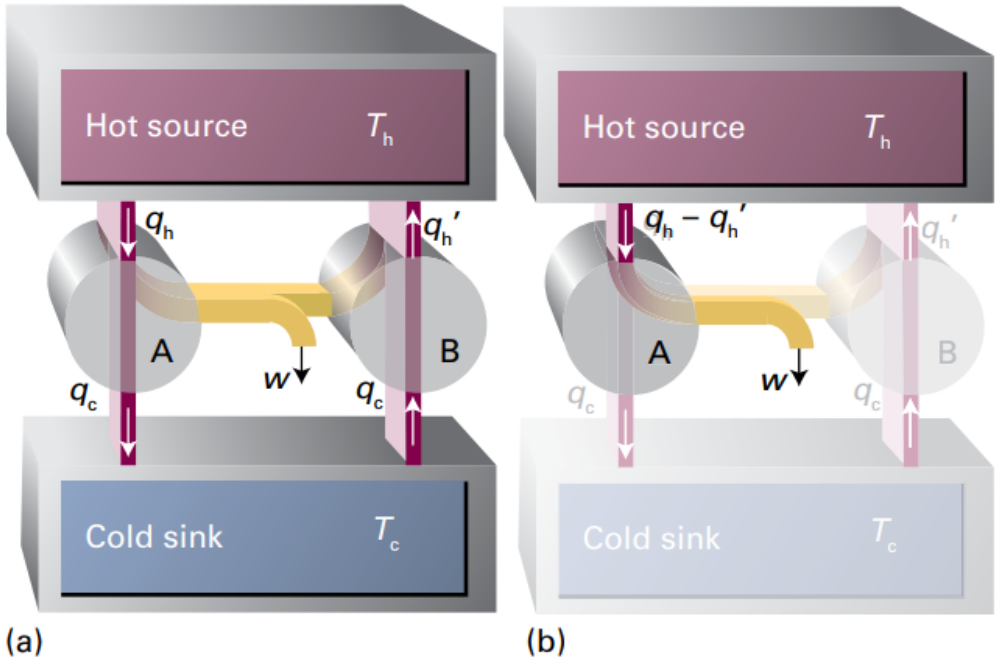


Figure 3A.11 (a) The demonstration of the equivalence of the efficiencies of all reversible engines working between the same thermal reservoirs is based on the flow of energy represented in this diagram. (b) The net effect of the processes is the conversion of heat into work without there being a need for a cold sink. This is contrary to the Kelvin statement of the Second Law.

S is a state function

- Any reversible cycle can be approximated as a collection of Carnot cycles
- Carnot cycles A, B, and C fitted together in such a way that their perimeter approximates the cycle indicated by the purple line.
- The path shown by the purple line can be approximated more closely by using more Carnot cycles, each of which is much smaller, and in the limit that they are infinitesimally small their perimeter matches the purple path exactly.
- This result implies that dS is an exact differential and therefore that S is a state function

$$\oint \mathrm{d}S = \oint \frac{\mathrm{d}q_{\mathrm{rev}}}{T} = 0 \qquad (3A.5)$$

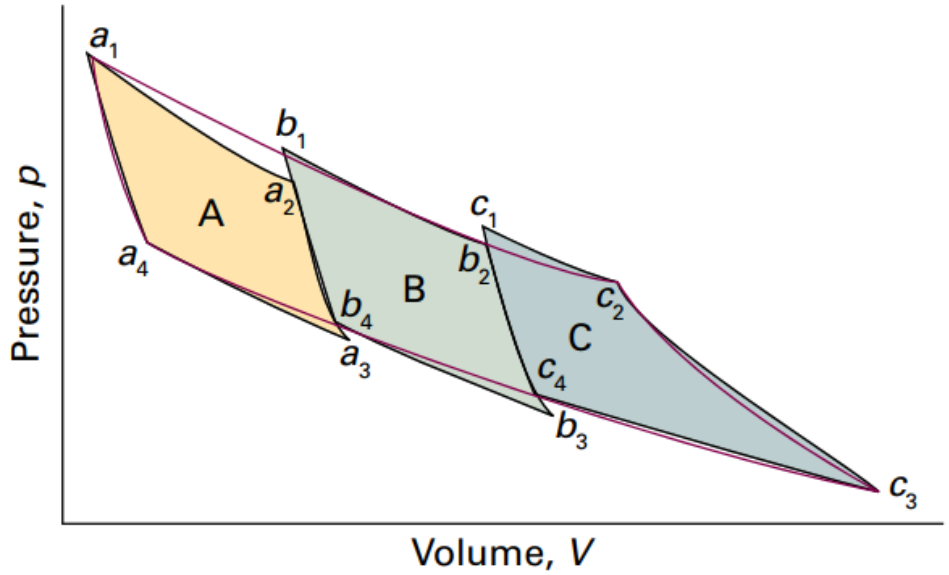


Figure 3A.12 The path indicated by the purple line can be approximated by traversing the *overall perimeter* of the area created by the three Carnot cycles A, B, and C; for each individual cycle the overall entropy change is zero. The entropy changes along the adiabatic segments (such as a_1–a_4 and c_2–c_3) are zero, so it follows that the entropy changes along the isothermal segments of any one cycle (such as a_1–a_2 and a_3–a_4) cancel. The entropy change resulting from traversing the overall perimeter of the three cycles is therefore zero.

The thermodynamic temperature

- Suppose an engine works reversibly between a hot source at atemperature T_h and a cold sink at a temperature T

$$T = (1 - \eta)T_h$$

- Kelvin to define the thermodynamic temperature scale in terms of the efficiency of a heat engine: construct an engine in which the hot source is at a known temperature and the cold sink is the object of interest. The temperature of the latter can then be inferred from the measured efficiency of the engine.
- The Kelvin scale (which is a special case of the thermodynamic temperature scale) is currently defined by using water at its triple point三相点[固、液、气三相可以同时平衡共存] as the notional hot source and defining that temperature as 273.16 K exactly

The Clausius inequality

- more work is done when a change is reversible than when it is irreversible $|dw_{rev}| \geq |dw|$
- dw and dw_{rev} are negative when energy leaves the system as work, $-dw_{rev} \geq -dw$, $dw - dw_{rev} \geq 0$
- The internal energy is a state function, so its change is the same for irreversible and reversible paths between the same two states $dU = dq + dw = dq_{rev} + dw_{rev}$, $dq_{rev} - dq \geq 0$ and therefore $dq_{rev} \geq dq$
- dqrev/T ≥ dq/T, the thermodynamic definition of the entropy ($dS = dq_{rev}/T$)

$$dS \geq \frac{dq}{T}$$ 克劳修斯不等式 Clausius inequality (3A.11)

- In an isolated system the entropy cannot decrease when a spontaneous change occurs, dq=0, dS>0
- The Clausius inequality also implies that spontaneous processes are also necessarily irreversible processes

$$dS_{tot} = \overbrace{dS}^{\geq dq/T} + \overbrace{dS_{sur}}^{-dq/T} \geq 0$$

inequality corresponds to an irreversible process
and the equality to a reversible process

- A spontaneous process ($dS_{tot} > 0$) is an irreversible process. A reversible process, for which $dS_{tot} = 0$, is spontaneous in neither direction: it is at equilibrium.

$$dS \geq \frac{dq}{T}$$ 克劳修斯不等式 Clausius inequality (3A.11)

The Clausius inequality

- The cooling of an object to the temperature of its surroundings, is indeed spontaneous
- Consider the transfer of energy as heat from one system—the hot source—at a temperature Th to another system—the cold sink—at a temperature Tc
- |dq| leaves the hot source (so $dq_h < 0$), the Clausius inequality implies that $dS \geq dq_h/T_h$. When |dq| enters the cold sink the Clausius inequality implies that $dS \geq dq_c/T_c$ (with $dq_c > 0$)

- $$dS \geq \frac{dq_h}{T_h} + \frac{dq_c}{T_c}$$

However, $dq_h = -dq_c$, so

$$dS \geq -\frac{dq_c}{T_h} + \frac{dq_c}{T_c} = \left(\frac{1}{T_c} - \frac{1}{T_h}\right) dq_c$$

which is positive (because $dq_c > 0$ and $T_h \geq T_c$).

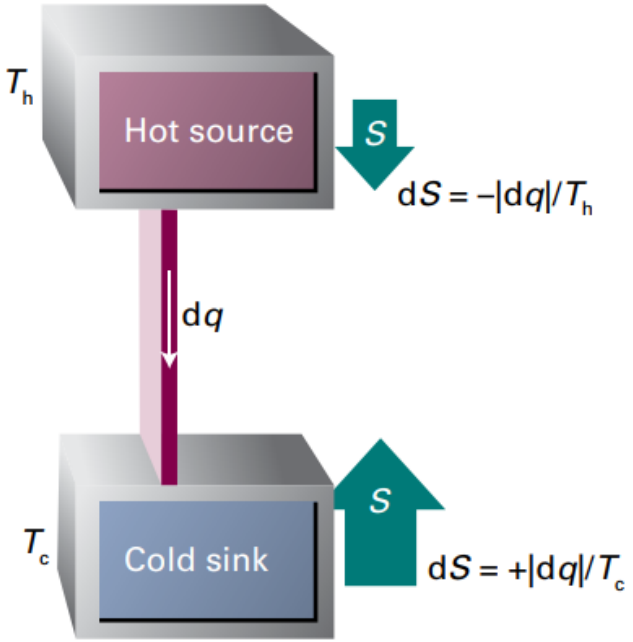


Figure 3A.13 When energy leaves a hot source as heat, the entropy of the source decreases. When the same quantity of energy enters a cooler sink, the increase in entropy is greater. Hence, overall there is an increase in entropy and the process is spontaneous. Relative changes in entropy are indicated by the sizes of the arrows.

TOPIC 3A · SELF-TEST

Check your understanding

For a spontaneous process in an isolated system, ΔS_{total} is:

- A. negative
- B. zero
- C. positive
- D. undefined

Answer · 答案: C — the total entropy increases for a spontaneous process.

$\Delta S_{total} = 0$ only at equilibrium.

Checklist of concepts

- ☐ 1. The **entropy** is a signpost of spontaneous change: the entropy of the universe increases in a spontaneous process.
- ☐ 2. A change in entropy is defined in terms of reversible heat transactions.
- ☐ 3. The **Boltzmann formula** defines entropy in terms of the number of ways that the molecules can be arranged amongst the energy states, subject to the arrangements having the same overall energy.
- ☐ 4. The **Carnot cycle** is used to prove that entropy is a state function.
- ☐ 5. The **efficiency** of a heat engine is the basis of the definition of the thermodynamic temperature scale and one realization of such a scale, the Kelvin scale.
- ☐ 6. The **Clausius inequality** is used to show that the entropy of an isolated system increases in a spontaneous change and therefore that the Clausius definition is consistent with the Second Law.
- ☐ 7. Spontaneous processes are irreversible processes; processes accompanied by no change in entropy are at equilibrium.

Entropy is the arrow of spontaneous change.

Checklist of equations

Property	Equation	Comment	Equation number
Thermodynamic entropy	$dS = dq_{rev}/T$	Definition	3A.1a
Entropy change of surroundings	$\Delta S_{sur} = q_{sur}/T_{sur}$		3A.2b
Boltzmann formula	$S = k \ln \mathcal{W}$	Definition	3A.4
Carnot efficiency	$\eta = 1 - T_c/T_h$	Reversible processes	3A.9
Thermodynamic temperature	$T = (1 - \eta)T_h$		3A.10
Clausius inequality	$dS \geq dq/T$		3A.11

Entropy is the arrow of spontaneous change.